Section 5: Solids, liquids and gases

a) Units

b) Density and pressure

c) Change of state

d) Ideal gas molecules

a) Units

Students will be assessed on their ability to:

5.1 use the following units: degrees Celsius (^{o}C), kelvin (K), joule (J), kilogram (kg), kilogram/metre^3 (kg/m^3), metre (m), metre^2 (m^2), metre^3 (m^3), metre/second (m/s), metre/second^2 (m/s^2), newton (N), pascal (Pa).

b) Density and pressure

Students will be assessed on their ability to:

5.2 know and use the relationship between density, mass and volume:

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

$$\rho = \frac{m}{V}$$

5.3 describe experiments to determine density using direct measurements of mass and volume

5.4 know and use the relationship between pressure, force and area:

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

$$p = \frac{F}{A}$$

5.5 understand that the pressure at a point in a gas or liquid which is at rest acts equally in all directions

5.6 know and use the relationship for pressure difference:

pressure difference = height × density × g

$$p = h \times \rho \times g$$

c) Change of state

Students will be assessed on their ability to:

5.7 understand the changes that occur when a solid melts to form a liquid, and when a liquid evaporates or boils to form a gas

5.8 describe the arrangement and motion of particles in solids, liquids and gases

d) Ideal gas molecules

Students will be assessed on their ability to:

5.9 understand the significance of Brownian motion, as supporting evidence for particle theory

5.10 understand that molecules in a gas have a random motion and that they exert a force and hence a pressure on the walls of the container

5.11 understand why there is an absolute zero of temperature which is –273°C

5.12 describe the Kelvin scale of temperature and be able to convert between the Kelvin and Celsius scales

5.13 understand that an increase in temperature results in an increase in the average speed of gas molecules

5.14 understand that the Kelvin temperature of the gas is proportional to the average kinetic energy of its molecules

5.15 describe the qualitative relationship between pressure and Kelvin temperature for a gas in a sealed container